

Dynamics accelerate the kinetics of ion diffusion through channels:
continuous-time random walk models beyond the mean field
approximation

Supplementary Information

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Dependence of D_{VACF} on Site Size

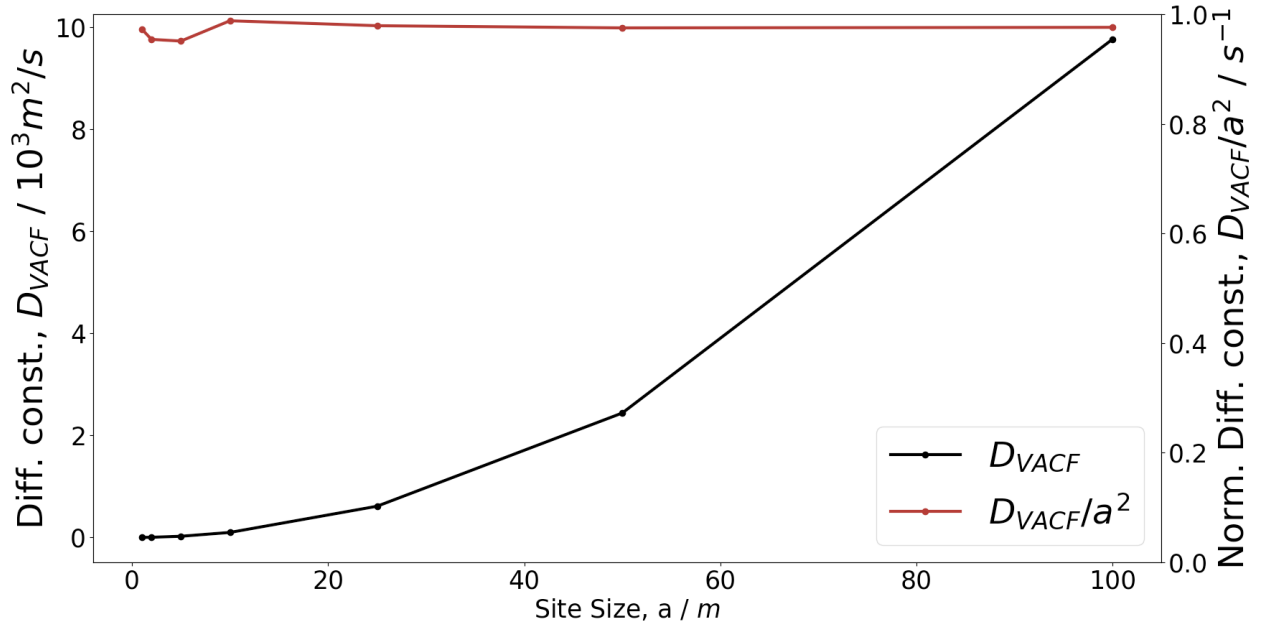


Figure S1: Plot of D_{VACF} and D_{VACF}/a^2 against site size (a) for a 2D lattice with periodic boundary conditions. Each simulation is run for 500,000 steps and averaged over 5000 trajectories. D_{VACF} scales as a^2 , as expected from Equation 8.

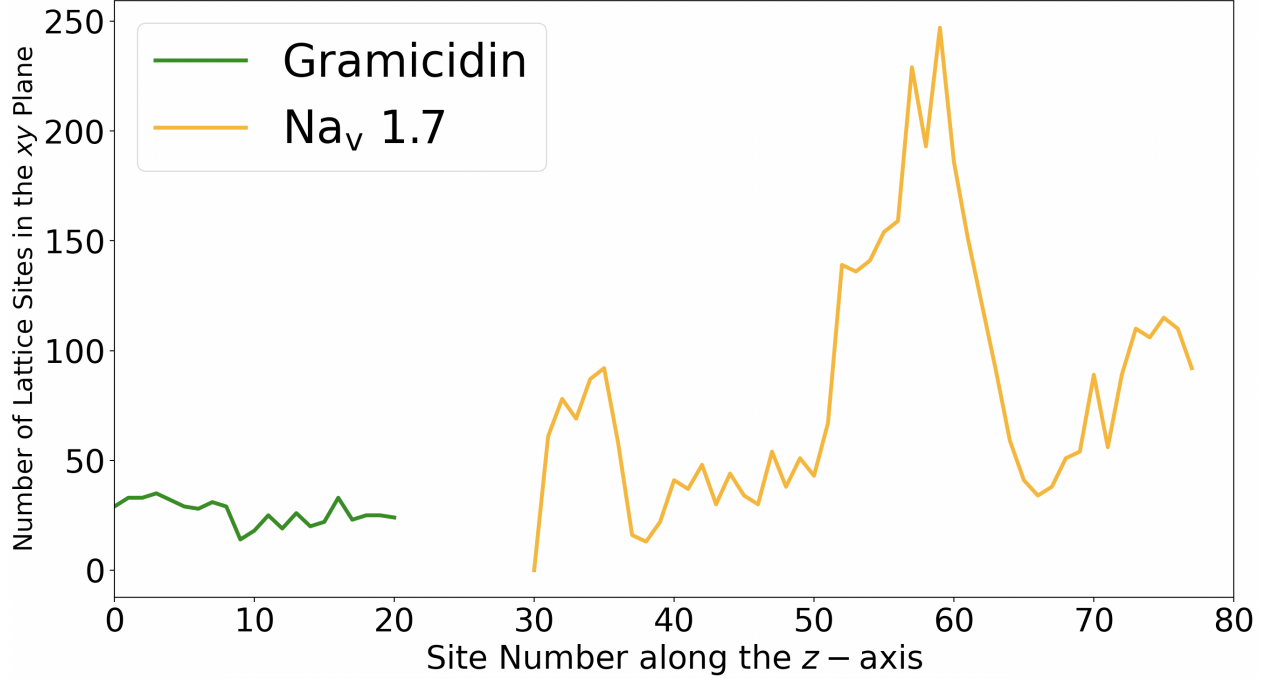


Figure S2: Number of sites in the xy -plane of the Gramicidin and $\text{Na}_v 1.7$ lattices as a function of the site number along the z -axis. The open lattice has a 4489 sites in the xy -plane throughout ($= 67 \times 67$). The site number has been kept consistent with the values in Table I.

Lattice Size

Rate distributions

Table S1: Means ($\mu_{\mathcal{D}^{LN}}$ and $\mu_{\mathcal{D}^G}$) and standard deviations ($\sigma_{\mathcal{D}^{LN}}$ and $\sigma_{\mathcal{D}^G}$) of the two distributions used in this work ($\mathcal{D}^{LN}$ and $\mathcal{D}^G$) against the desired means and standard deviations (k_{ref} and σ). The values are calculated by randomly drawing 100,000 numbers from the respective distributions.

(k_{ref}, σ)	$(\mu_{\mathcal{D}^{LN}}, \sigma_{\mathcal{D}^{LN}})$	$(\mu_{\mathcal{D}^G}, \sigma_{\mathcal{D}^G})$
(30000,300)	(29999,300)	(30000,301)
(30000,3000)	(30023,3008)	(29995,2995)
(30000,7500)	(29999,7479)	(29975,7536)
(30000,15000)	(30092,15048)	(30099,14053)
(30000,22500)	(30019,22542)	(30709,18330)
(30000,30000)	(30068,30223)	(30185,20877)

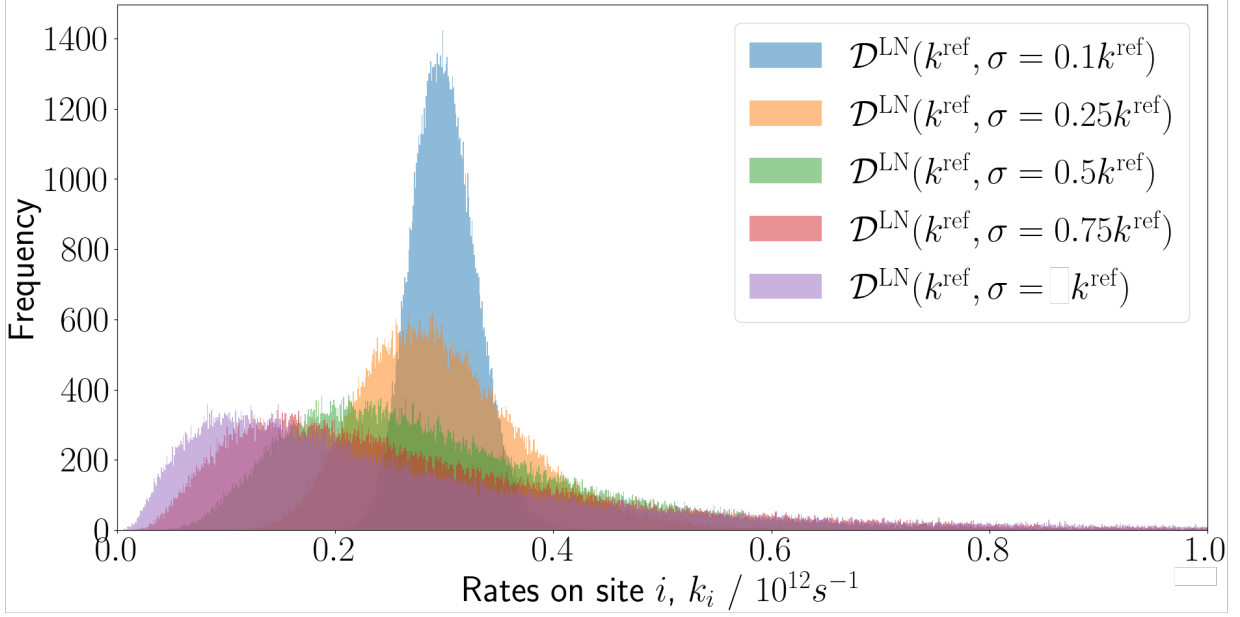


Figure S3: Distribution $\mathcal{D}^{\text{LN}}(k^{\text{ref}}, \sigma)$ at various σ . $\mathcal{D}^{\text{LN}}$ is the $\mathcal{D}$ distribution realized from a log-normal distribution $\mathcal{L}_N(\mu_{\mathcal{L}_N}, \sigma_{\mathcal{L}_N})$ defined in the Methods section and using the underlying gaussian distribution.

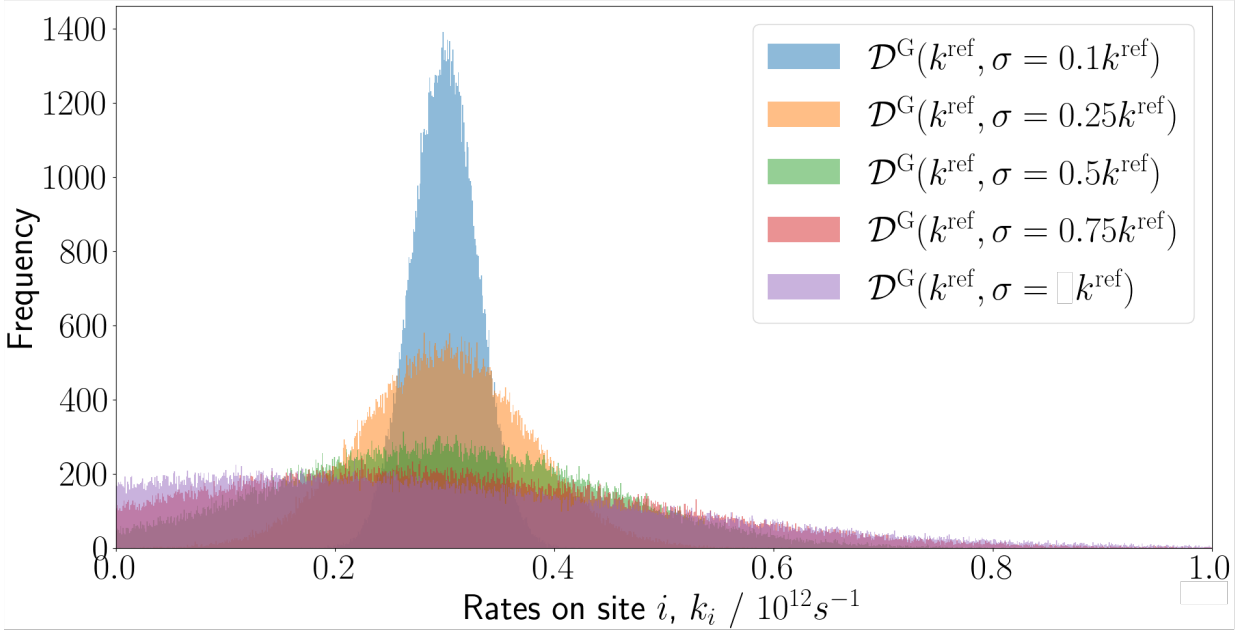


Figure S4: Distribution $\mathcal{D}^{\text{G}}(k^{\text{ref}}, \sigma)$ at various σ . $\mathcal{D}^{\text{G}}$ is the $\mathcal{D}$ distribution realized from a gaussian distribution $\mathcal{G}(\mu_{\mathcal{G}}, \sigma_{\mathcal{G}})$ defined in the Methods section and re-drawing from $\mathcal{G}$ until a positive value is obtained.

Comparison of the VACF and MSD approaches to the computation of the diffusion coefficient

We find that D_{VACF} converges faster and over fewer trajectories than the MSD. Figure S5 shows the MSD and the progression of D_{VACF} for a random walk without bias or disorder in the open and $\text{Na}_v1.7$ lattices.

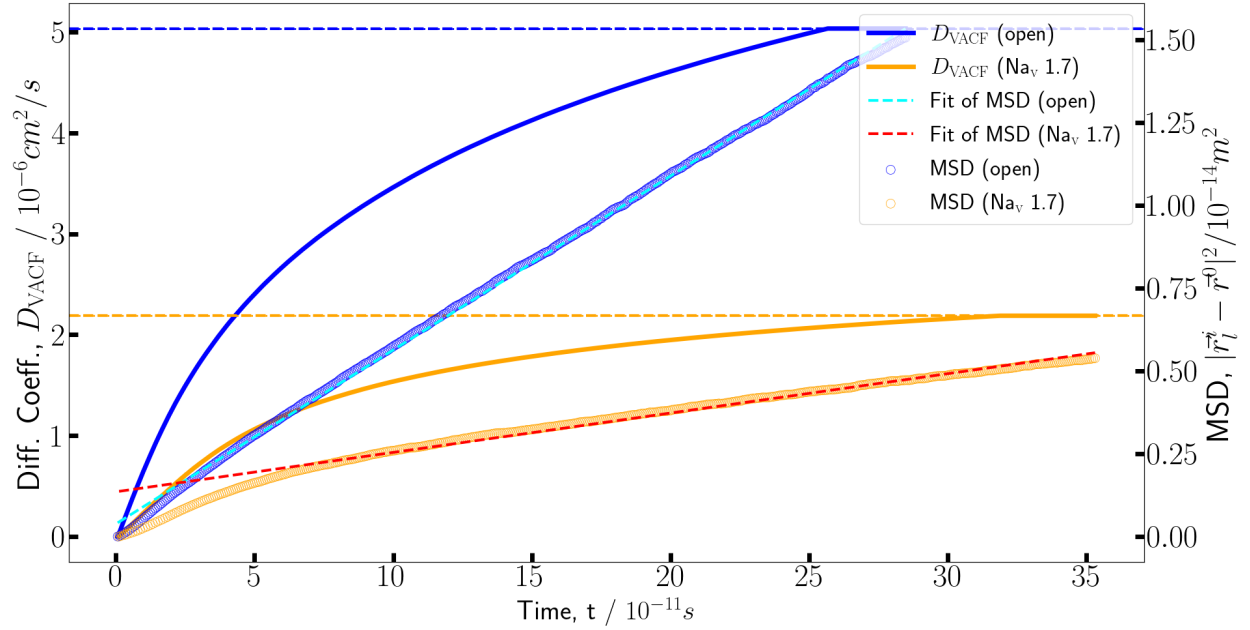


Figure S5: D_{VACF} (left axis) and MSD (right axis) as a function of time for unbiased, undisordered, walks in the open and $\text{Na}_v1.7$ lattices (averaged of 5000 simulations of 500 steps). In both cases, a linear fit to the MSD is shown as a dotted line. The slope of the fitted line gives $D_{\text{MSD}} = 5.252 \times 10^{-6}$ and $1.188 \times 10^{-6} \text{ cm}^2.\text{s}^{-1}$ for the open and $\text{Na}_v1.7$ lattice, respectively. $D_{\text{VACF}} = 5.03 \times 10^{-6}$ and $2.2 \times 10^{-6} \text{ cm}^2.\text{s}^{-1}$ for the open and $\text{Na}_v1.7$ lattice, respectively.

We find a good fit ($R^2 > 0.9$) for both cases over the last 90% of the trajectory. In this case, both D_{VACF} and D_{MSD} yield values close to the theoretical value of D^{exp} . In contrast, Figure S6 shows the MSD and the progression of D_{VACF} for a random walk without disorder but with bias in the open and $\text{Na}_v1.7$ lattices. In this case, the MSD is not linear, and no reasonable fit can be used to estimate the value of D_{MSD} . However, D_{VACF} converges within the walk parameters.

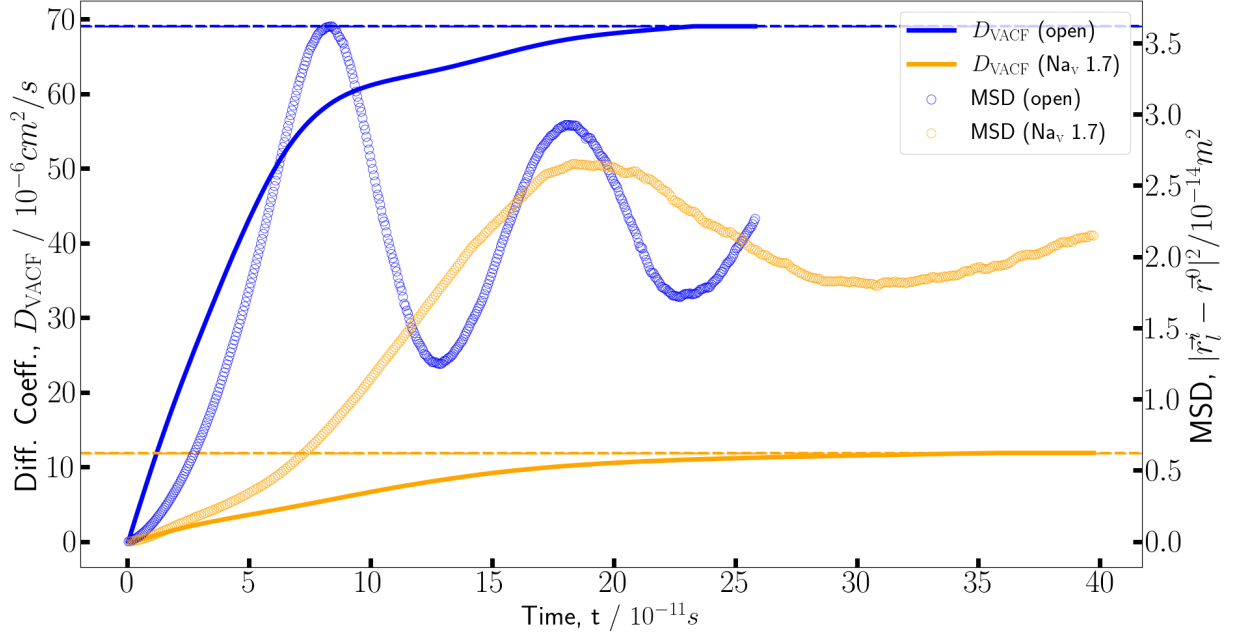


Figure S6: D_{VACF} (left axis) and MSD (right axis) as a function of time for biased, undisordered, walks in the open and $\text{Na}_v1.7$ lattices (averaged of 5000 simulations of 500 steps). $D_{\text{VACF}} = 69.0 \times 10^{-6}$ and $11.9 \times 10^{-6} \text{cm}^2.\text{s}^{-1}$ for the open and $\text{Na}_v1.7$ lattice, respectively.

Raw data for the graphs plotted in the results section of the manuscript

Table S2: D_{VACF} shown as “mean (95% confidence interval)” (over 5 sets of 5000 CTRW walks of 500 steps each) for the three lattices with static disorder. Rates are drawn from $\mathcal{D}^{\text{LN}}$.

b_{-z}	b_{+z}	σ in k^{ref}	D_{VACF} (Open Lattice) in $10^{-6} \text{ cm}^2.\text{s}^{-1}$	D_{VACF} (Gramicidin) in $10^{-6} \text{ cm}^2.\text{s}^{-1}$	D_{VACF} ($\text{Na}_v1.7$) in $10^{-6} \text{ cm}^2.\text{s}^{-1}$
1	1	0	5.1 (5.1,5.1)	2.1 (2.0,2.2)	1.8 (1.8,1.8)
1	1	0.01	5.1 (5.0,5.2)	2.1 (2.0,2.2)	1.8 (1.8,1.8)
1	1	0.1	5.0 (4.9,5.1)	2.1 (2.0,2.2)	1.8 (1.8,1.8)
1	1	0.25	4.9 (4.8,5.0)	2.0 (1.9,2.1)	1.7 (1.6,1.8)
1	1	0.5	4.6 (4.3,4.9)	2.0 (1.3,2.7)	1.6 (1.2,2.0)
1	1	0.75	4.1 (3.9,4.3)	1.8 (1.5,2.1)	1.5 (1.1,1.9)
1	1	1	3.8 (3.7,3.9)	1.9 (1.5,2.3)	1.4 (1.1,1.7)
2	1/2	0	68.5 (68.0,69.0)	41.1 (40.0,42.2)	11.9 (11.6,12.2)
2	1/2	0.01	68.6 (67.3,69.9)	41.0 (38.6,43.4)	11.9 (11.5,12.3)
2	1/2	0.1	67.7 (66.3,69.1)	40.2 (37.1,43.3)	11.9 (11.2,12.6)
2	1/2	0.25	65.0 (60.7,69.3)	40.8 (33.1,48.5)	11.0 (10.6,11.4)
2	1/2	0.5	55.5 (53.0,58.0)	34.5 (21.2,47.8)	9.7 (8.4,11.0)
2	1/2	0.75	47.3 (40.3,54.3)	22.5 (10.9,34.1)	7.6 (5.8,9.4)
2	1/2	1	35.5 (29.3,41.7)	15.3 (9.8,20.8)	6.7 (4.6,8.8)

Table S3: D_{VACF} shown as “mean (95% confidence interval)” (over 5 sets of 5000 CTRW walks of 500 steps each) for the three lattices with dynamic disorder. Rates are drawn from $\mathcal{D}^{\text{LN}}$.

b_{-z}	b_{+z}	σ in k^{ref}	D_{VACF} (Open Lattice) in $10^{-6} \text{ cm}^2 \cdot \text{s}^{-1}$	D_{VACF} (Gramicidin) in $10^{-6} \text{ cm}^2 \cdot \text{s}^{-1}$	D_{VACF} (Na_v1.7) in $10^{-6} \text{ cm}^2 \cdot \text{s}^{-1}$
1	1	0	5.1 (5.0,5.2)	2.2 (2.1,2.3)	1.8 (1.8,1.8)
1	1	0.01	5.1 (5.0,5.2)	2.2 (2.1,2.3)	1.8 (1.8,1.8)
1	1	0.1	5.1 (5.0,5.2)	2.1 (2.0,2.2)	1.8 (1.8,1.8)
1	1	0.25	5.0 (5.0,5.0)	2.1 (2.0,2.2)	1.7 (1.7,1.7)
1	1	0.5	4.7 (4.6,4.8)	2.0 (1.9,2.1)	1.6 (1.5,1.7)
1	1	0.75	4.3 (4.2,4.4)	1.7 (1.6,1.8)	1.4 (1.4,1.4)
1	1	1	3.8 (3.7,3.9)	1.5 (1.5,1.5)	1.2 (1.2,1.2)
2	1/2	0	68.6 (67.9,69.3)	39.0 (37.0,41.0)	11.9 (11.5,12.3)
2	1/2	0.01	68.7 (67.5,69.9)	39.9 (37.8,42.0)	11.9 (11.6,12.2)
2	1/2	0.1	68.2 (66.8,69.6)	39.5 (36.3,42.7)	11.9 (11.4,12.4)
2	1/2	0.25	68.3 (66.9,69.7)	39.9 (38.1,41.7)	11.7 (11.3,12.1)
2	1/2	0.5	68.0 (66.5,69.5)	38.1 (36.0,40.2)	10.8 (10.6,11.0)
2	1/2	0.75	65.7 (64.5,66.9)	36.9 (34.4,39.4)	9.6 (9.4,9.8)
2	1/2	1	62.3 (61.1,63.5)	34.9 (33.5,36.3)	8.4 (8.2,8.6)

Table S4: D_{VACF} shown as “mean (95% confidence interval)” (over 5 sets of 5000 CTRW walks of 500 steps each) for the three lattices with static disorder. Rates are drawn from $\mathcal{D}^{\text{G}}$.

b_{-z}	b_{+z}	σ in k^{ref}	D_{VACF} (Open Lattice) in $10^{-6} \text{ cm}^2 \cdot \text{s}^{-1}$	D_{VACF} (Gramicidin) in $10^{-6} \text{ cm}^2 \cdot \text{s}^{-1}$	D_{VACF} (Na_v1.7) in $10^{-6} \text{ cm}^2 \cdot \text{s}^{-1}$
1	1	0	5.1 (5.0,5.2)	2.1 (2.1,2.2)	1.8 (1.7,1.8)
1	1	0.01	5.1 (4.9,5.2)	2.1 (2.0,2.3)	1.8 (1.7,1.8)
1	1	0.1	5.1 (5.0,5.2)	2.1 (2.0,2.2)	1.8 (1.7,1.8)
1	1	0.25	4.9 (4.9,5.0)	2.1 (2.0,2.2)	1.7 (1.6,1.9)
1	1	0.5	4.7 (4.4,5.0)	2.1 (1.5,2.7)	1.6 (1.3,1.9)
1	1	0.75	4.6 (4.1,5.1)	2.0 (1.3,2.7)	1.5 (1.0,2.0)
1	1	1	4.1 (3.6,4.7)	1.8 (1.4,2.3)	1.4 (1.1,1.6)
2	1/2	0	68.5 (66.8,70.2)	41.2 (39.0,43.4)	12.0 (11.8,12.1)
2	1/2	0.01	68.4 (66.8,69.9)	41.3 (38.6,44.0)	12.0 (11.5,12.5)
2	1/2	0.1	68.2 (67.3,69.1)	40.0 (37.7,42.4)	11.7 (11.1,12.4)
2	1/2	0.25	65.1 (63.4,66.8)	36.6 (24.2,49.0)	11.4 (9.3,13.4)
2	1/2	0.5	54.4 (50.0,58.9)	27.4 (18.6,36.2)	8.8 (6.8,10.9)
2	1/2	0.75	52.0 (44.6,59.4)	28.7 (6.6,50.9)	8.2 (6.5,9.8)
2	1/2	1	44.3 (41.3,47.2)	16.7 (-6.8,40.2)	8.0 (4.6,11.3)

Table S5: D_{VACF} shown as “mean (95% confidence interval)” (over 5 sets of 5000 CTRW walks of 500 steps each) for the three lattices with dynamic disorder. Rates are drawn from $\mathcal{D}^G$.

b_{-z}	b_{+z}	σ in k^{ref}	D_{VACF} (Open Lattice) in $10^{-6} \text{ cm}^2 \cdot \text{s}^{-1}$	D_{VACF} (Gramicidin) in $10^{-6} \text{ cm}^2 \cdot \text{s}^{-1}$	D_{VACF} (Na_v1.7) in $10^{-6} \text{ cm}^2 \cdot \text{s}^{-1}$
1	1	0	5.1 (5.0,5.2)	2.2 (2.0,2.3)	1.7 (1.7,1.8)
1	1	0.01	5.1 (5.0,5.2)	2.2 (2.0,2.3)	1.8 (1.8,1.8)
1	1	0.1	5.0 (4.9,5.2)	2.2 (2.1,2.2)	1.8 (1.7,1.8)
1	1	0.25	4.9 (4.8,5.0)	2.1 (2.0,2.2)	1.7 (1.6,1.8)
1	1	0.5	4.7 (4.6,4.8)	1.9 (1.8,2.0)	1.6 (1.5,1.6)
1	1	0.75	4.7 (4.6,4.8)	1.8 (1.7,2.0)	1.5 (1.4,1.5)
1	1	1	4.3 (4.2,4.3)	1.7 (1.6,1.7)	1.3 (1.3,1.4)
2	1/2	0	68.5 (67.4,69.6)	38.2 (35.3,41.0)	11.7 (11.1,12.4)
2	1/2	0.01	68.5 (67.4,69.5)	39.2 (36.9,41.5)	12.0 (11.6,12.3)
2	1/2	0.1	68.9 (66.8,70.9)	39.1 (36.6,41.5)	11.8 (11.5,12.1)
2	1/2	0.25	68.3 (67.1,69.5)	39.4 (35.8,42.9)	11.5 (11.0,12.0)
2	1/2	0.5	66.3 (65.8,66.8)	35.7 (34.7,36.6)	10.4 (9.9,10.8)
2	1/2	0.75	69.2 (68.0,70.4)	38.6 (37.6,39.6)	10.0 (9.7,10.2)
2	1/2	1	-	-	-

Table S6: D_{VACF} for the sections of the Na_v1.7 pore, shown as “mean (95% confidence interval)” (over 5 sets of 5000 CTRW walks of 500 steps each) with static disorder. Rates are drawn from $\mathcal{D}^{\text{LN}}$.

b_{-z}	b_{+z}	σ in k^{ref}	D_{VACF} (Selectivity Filter) in $10^{-6} \text{ cm}^2 \cdot \text{s}^{-1}$	D_{VACF} (Central Cavity) in $10^{-6} \text{ cm}^2 \cdot \text{s}^{-1}$	D_{VACF} (Inactivation Gate) in $10^{-6} \text{ cm}^2 \cdot \text{s}^{-1}$
1	1	0	0.9 (0.9,0.9)	1.6 (1.5,1.7)	3.1 (2.6,3.5)
1	1	0.01	0.9 (0.9,0.9)	1.6 (1.5,1.7)	3.0 (2.3,3.6)
1	1	0.1	0.9 (0.9,0.9)	1.6 (1.5,1.7)	2.9 (2.4,3.3)
1	1	0.25	0.9 (0.8,0.9)	1.5 (1.4,1.7)	3.1 (2.3,3.8)
1	1	0.5	0.9 (0.7,1.0)	1.5 (1.3,1.7)	3.0 (2.3,3.7)
1	1	0.75	0.8 (0.6,1.0)	1.4 (1.2,1.5)	2.8 (2.2,3.5)
1	1	1	0.8 (0.6,0.9)	1.2 (0.9,1.5)	2.7 (1.4,4.0)
2	1/2	0	8.8 (7.4,10.1)	12.4 (12.0,12.8)	11.3 (11.0,11.6)
2	1/2	0.01	7.9 (3.7,12.0)	12.4 (11.7,13.1)	11.4 (11.1,11.8)
2	1/2	0.1	8.8 (3.7,14.0)	12.4 (11.8,13.1)	11.4 (10.7,12.1)
2	1/2	0.25	7.7 (5.3,10.1)	11.5 (11.1,11.9)	10.6 (10.1,11.1)
2	1/2	0.5	6.5 (3.3,9.8)	10.1 (8.9,11.4)	9.0 (7.8,10.2)
2	1/2	0.75	5.3 (3.8,6.7)	7.8 (6.3,9.4)	7.2 (6.2,8.2)
2	1/2	1	4.6 (2.7,6.5)	7.0 (4.9,9.0)	6.3 (5.1,7.5)

Table S7: D_{VACF} for the sections of the $\text{Na}_v1.7$ pore, shown as “mean (95% confidence interval)” (over 5 sets of 5000 CTRW walks of 500 steps each) with dynamic disorder. Rates are drawn from $\mathcal{D}^{\text{LN}}$.

b_{-z}	b_{+z}	σ in k^{ref}	D_{VACF} (Selectivity Filter) in $10^{-6} \text{ cm}^2 \cdot \text{s}^{-1}$	D_{VACF} (Central Cavity) in $10^{-6} \text{ cm}^2 \cdot \text{s}^{-1}$	D_{VACF} (Inactivation Gate) in $10^{-6} \text{ cm}^2 \cdot \text{s}^{-1}$
1	1	0	0.9 (0.8,0.9)	1.6 (1.5,1.7)	3.0 (2.5,3.5)
1	1	0.01	0.9 (0.9,0.9)	1.6 (1.5,1.7)	3.0 (2.1,3.9)
1	1	0.1	0.9 (0.8,0.9)	1.6 (1.4,1.7)	3.0 (2.4,3.6)
1	1	0.25	0.9 (0.8,0.9)	1.5 (1.5,1.6)	2.7 (2.4,3.0)
1	1	0.5	0.8 (0.8,0.8)	1.4 (1.3,1.5)	2.7 (2.5,2.8)
1	1	0.75	0.7 (0.6,0.7)	1.2 (1.2,1.3)	2.3 (2.1,2.5)
1	1	1	0.6 (0.6,0.6)	1.1 (1.0,1.1)	1.9 (1.5,2.3)
2	1/2	0	8.8 (7.1,10.6)	12.4 (12.1,12.6)	11.3 (11.2,11.5)
2	1/2	0.01	8.4 (5.0,11.8)	12.4 (12.0,12.8)	11.3 (11.1,11.5)
2	1/2	0.1	9.0 (6.9,11.2)	12.4 (11.9,13.0)	11.3 (11.0,11.6)
2	1/2	0.25	8.7 (5.7,11.8)	12.1 (11.7,12.5)	11.1 (10.8,11.5)
2	1/2	0.5	8.1 (4.4,11.8)	11.3 (10.7,12.0)	10.4 (10.0,10.8)
2	1/2	0.75	7.1 (3.2,11.0)	10.1 (9.8,10.3)	9.2 (9.1,9.3)
2	1/2	1	6.3 (2.2,10.3)	8.8 (8.5,9.2)	8.0 (7.7,8.4)

Table S8: D_{VACF} for the sections of the $\text{Na}_v1.7$ pore, shown as “mean (95% confidence interval)” (over 5 sets of 5000 CTRW walks of 500 steps each) with static disorder. Rates are drawn from $\mathcal{D}^{\text{G}}$.

b_{-z}	b_{+z}	σ in k^{ref}	D_{VACF} (Selectivity Filter) in $10^{-6} \text{ cm}^2 \cdot \text{s}^{-1}$	D_{VACF} (Central Cavity) in $10^{-6} \text{ cm}^2 \cdot \text{s}^{-1}$	D_{VACF} (Inactivation Gate) in $10^{-6} \text{ cm}^2 \cdot \text{s}^{-1}$
1	1	0	0.9 (0.8,0.9)	1.6 (1.5,1.7)	3.1 (2.5,3.6)
1	1	0.01	0.9 (0.9,0.9)	1.6 (1.5,1.6)	3.0 (2.6,3.3)
1	1	0.1	0.9 (0.9,0.9)	1.6 (1.5,1.7)	3.0 (2.4,3.5)
1	1	0.25	0.9 (0.8,0.9)	1.6 (1.4,1.7)	2.9 (2.4,3.4)
1	1	0.5	0.8 (0.7,0.9)	1.4 (1.2,1.6)	3.0 (2.2,3.7)
1	1	0.75	0.8 (0.7,1.0)	1.4 (1.3,1.6)	3.0 (1.8,4.2)
1	1	1	0.8 (0.6,1.0)	1.3 (1.1,1.5)	2.5 (1.6,3.4)
2	1/2	0	8.1 (4.5,11.7)	12.4 (11.9,12.9)	11.4 (10.9,12.0)
2	1/2	0.01	9.1 (7.8,10.4)	12.5 (12.0,13.1)	11.5 (11.2,11.7)
2	1/2	0.1	8.4 (4.7,12.1)	12.2 (11.6,12.7)	11.2 (10.7,11.7)
2	1/2	0.25	8.9 (5.9,11.9)	11.9 (10.3,13.6)	10.7 (10.0,11.3)
2	1/2	0.5	7.1 (2.2,12.0)	9.3 (7.0,11.6)	8.3 (7.2,9.4)
2	1/2	0.75	5.2 (3.5,6.9)	8.5 (7.6,9.5)	7.6 (6.5,8.7)
2	1/2	1	5.4 (2.0,8.7)	8.5 (5.6,11.5)	7.2 (4.8,9.5)

Table S9: D_{VACF} for the sections of the $\text{Na}_v1.7$ pore, shown as “mean (95% confidence interval)” (over 5 sets of 5000 CTRW walks of 500 steps each) with dynamic disorder. Rates are drawn from $\mathcal{D}^G$.

b_{-z}	b_{+z}	σ in k^{ref}	D_{VACF} (Selectivity Filter) in $10^{-6} \text{ cm}^2.\text{s}^{-1}$	D_{VACF} (Central Cavity) in $10^{-6} \text{ cm}^2.\text{s}^{-1}$	D_{VACF} (Inactivation Gate) in $10^{-6} \text{ cm}^2.\text{s}^{-1}$
1	1	0	0.9 (0.8,0.9)	1.6 (1.5,1.6)	3.0 (2.2,3.9)
1	1	0.01	0.9 (0.9,0.9)	1.6 (1.6,1.7)	3.0 (2.5,3.4)
1	1	0.1	0.9 (0.9,0.9)	1.6 (1.4,1.7)	3.0 (2.5,3.4)
1	1	0.25	0.9 (0.8,0.9)	1.5 (1.4,1.7)	3.0 (2.6,3.3)
1	1	0.5	0.8 (0.8,0.8)	1.4 (1.4,1.5)	2.8 (2.2,3.4)
1	1	0.75	0.7 (0.7,0.8)	1.3 (1.2,1.4)	2.6 (2.1,3.1)
1	1	1	0.7 (0.7,0.7)	1.2 (1.1,1.2)	2.2 (1.8,2.6)
2	1/2	0	8.8 (6.8,10.8)	12.3 (11.4,13.1)	11.2 (10.7,11.7)
2	1/2	0.01	8.3 (3.9,12.7)	12.6 (12.2,12.9)	11.4 (11.1,11.6)
2	1/2	0.1	9.2 (6.4,12.0)	12.3 (11.8,12.8)	11.3 (10.9,11.8)
2	1/2	0.25	8.5 (5.9,11.2)	12.0 (11.5,12.6)	10.9 (10.6,11.2)
2	1/2	0.5	7.2 (4.1,10.4)	10.8 (10.3,11.4)	9.8 (9.2,10.3)
2	1/2	0.75	8.2 (6.8,9.6)	10.5 (10.0,11.0)	9.6 (9.4,9.9)
2	1/2	1	-	-	-